

Representation Convergence — Results Summary

Every experiment's measured outcome in one place. Thesis: independently trained models share representations up to a linear transformation. All figures are held-out; controls and chance rates are quoted alongside. Full method and caveats in [Final_Project_Report.pdf](#) (61 pp).

The findings at a glance

#	finding	headline evidence
1	Convergence is linear	MLP = ridge to 4 decimals (0.9183 = 0.9183), reproduced on two independent runs of a fresh corpus
2	Related, not rigid	Procrustes R2 0.038 vs ridge 0.592; 69x spread in the map's singular values
3	Deployable	95% of ceiling at 0.79 MB; verification AUC 0.999
4	Cross-modal without joint training	DINOv2 -> bge 40.9-58.3% of a matched ceiling, up to 358x chance
5	Alignment scales with capacity	+15.3 points of ceiling per decade of parameters, six monotone measures
6	Isotropy, not similarity-training	whitening a raw LM space reaches 102.6% / 101.4% of a contrastive encoder at two scales
7	Correspondence is self-identifying	Gromov-Wasserstein 39.7% exact = 595x chance, zero paired examples
8	One shared space, used	hub transfer 92.9-96.5% of native on encoders never seen, incl. cross-lineage
9	The mechanism predicts	pre-registered encoder ladder, rank correlation +1.00

Meta-finding, measured three times. Isotropy helps when READING a correspondence (C.11), hurts when FINDING one (C.12), and hurts when CARRYING several at once (the hub width cliff). One geometric property, three opposite prescriptions.

Experiment A — scientific proof (GPT-2 vs Pythia-160M)

metric	fail	pass	measured
CKA diagonal mean (trained)	<0.15	>0.35	0.424
CKA mean, random control	>0.20	<0.15	0.067
trained / random ratio	<2x	>=3x	6.3x
stitching R2, peak layer (L11)	<0.50	>0.70	0.797
stitching R2, random control	>0.40	<0.30	0.173
depth-slope signature	both flat	opposite	0.59->0.80 vs 0.38->0.09

Verdict: PASS. The trained curve rises with depth while the random control decays - opposite slopes, an 8x gap at L11. Run 1 (1,646 samples) failed on sample starvation; run 2 (10,000) passed decisively, identical pipeline.

Experiment B — the deployed adapter (MobileCLIP-S1 -> SigLIP 2)

quantity	value
Ridge adapter: held-out R2 / mean cosine	0.592 / 0.900
Procrustes (rotation only): R2	0.038 (anisotropy, not just orientation)
MLP adapter: best cosine	0.903 (+0.003 -> relationship is linear)
Retrieval R@1 / R@5 / R@10 (ceiling)	0.630 / 0.885 / 0.944
Retrieval with linear adapter	0.557 / 0.843 / 0.918
as % of ceiling	88.4% / 95.3% / 97.2%
raw cross-space baseline	0.001 / 0.005 / 0.010 (exact chance)
verification / ranking / membership	AUC 0.999 / R@1 0.987 / AUC 0.889
hubness N10 skew: raw -> adapted	4.28 -> 1.61 (ceiling 1.41)
deployed artifact	Adapter.mlpackage, 0.79 MB fp16, iOS 16+

Series E — cross-modal, scaling, geometry

experiment	result
E1 capacity sweep (DINOv2 -> bge)	small/base/large R@1 40.9% / 53.5% / 58.3% of ceiling; +15.3 pts per decade, fit R2 0.949
E1.1 swap text encoder to GPT-2	R2 retains 71%, retrieval collapses to 34%; cosine 1.000 while R@1 0.160
E1.2 isotropy rescue	whitening lifts R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder; eff. rank 6.5 -> 490.8; iso-retrieval corr +0.94
E1.3 encoder ladder (pre-registered)	P1 rank correlation +1.00 (confirmed); P2 partial - isotropy necessary, not sufficient

Series F — the contrast condition (deliberate training)

experiment	result
F1/F2 capacity test	MLP converges to EXACTLY linear: cosine 0.9183 = 0.9183, gain +0.0000, reproduced on two runs from a fresh 38k-row corpus (74.2 rows/dim)
F5 data test	linear refit on 38k gives R@1 +0.0244 (bootstrap CI excludes 0); 90%-of-ceiling gate reported NOT met at 89.0%
F3 unfreeze bottleneck	built and gated shut - running it would cost the clean 'no pretrained weight modified' claim; the gate is the contribution

Series C — pair-free recovery

quantity	value
Gromov-Wasserstein exact matches	39.7% = 595x chance, zero pairs
map fitted from that assignment	R@1 0.252 (40% of ceiling), verification AUC 0.906
negative 1: ICP refinement	degrades monotonically (converges on its own errors)
negative 2: whitening before matching	39.7% -> 0.1% (destroys matching)
scope (C.12.1)	both encoders contrastive image-text - the WEAKEST-independence pair; result is bounded accordingly

Series G — the shared hub

experiment	result
G1 seven-space hub, transfer	R@1 0.451 at HUB_DIM 512 (AUC 0.986); transfer cliff at 512->768; vs 0.445 at five spaces - UNCHANGED (within noise), which is the useful result
G1 composition (scale-free R@1)	95.6% retention over 36 of 41 pairs; the 5 exceptions all write INTO GPT-2
G1 read/write asymmetry	GPT-2 as source 95.1%, as target 24.8% - a collapsed space is readable out of, not writable into; tracks effective rank across bge/BERT/GPT-2
G3 compatibility (vs bespoke hubs)	42 of 42 ordered pairs SHARED, mean retention 103.6%; control at chance (0.004 vs 0.001)
G4 cross-lineage transfer	SigLIP 2, an INDEPENDENT lineage, retains 94.2% of a natively-fitted head - inside the 93.8-96.5% within-family band; control at chance. Cross-lineage, not simultaneously cross-objective; still a ViT
G5 scale of agreement	CKNNA: 21/21 pairs higher locally than at the global limit; agreement is LOCAL. The raw ratio-to-chance curve is NOT the evidence - it is bounded by N/k (C.13.9)
G5 what explains it (exploratory)	objective (+53x) > lineage (+44x) > modality (+29x); top pair bge-SBERT shares objective, no lineage
G5 CKNNA (the literature's metric)	21 of 21 pairs score higher locally than at the global limit; rank correlation +0.81 with our own measure. The local-vs-global conclusion does not depend on the instrument
G5 permutation calibration	k-NN null = k/N exactly (already calibrated); CKA null +0.094 and RISES with width. Correcting it WIDENS the objective-over-lineage gap from +0.024 to +0.181 - the confound had favoured the lineage pair
G6 manifold characterisation	intrinsic dimension 11.2-19.9 inside 768-2048 wide spaces (<3% of ambient); Procrustes across 21 pairs mean 0.194 vs ridge 0.667; CCA shows cross-modal pairs share as few as 6 of 64 directions

G7 what the hub does to local structure	on healthy spaces NEUTRAL, and now BOUNDED (K3): cluster-bootstrap 95% CI -0.038 to +0.024, MDE 0.034 vs a pre-registered 0.050 - a demonstrated null, not absence of evidence. Six pairs from four encoders, so the interval resamples encoders. Rebuilt -0.010 vs original +0.010: no stable sign. Degenerate pairs +0.151 (C.11 rescue, 4th instance)
G7 the width cliff	reproduced on ZERO-SHOT transfer: 0.489 at 512 -> 0.312 at 768; self-transfer plateaus and cannot cliff. SIX explanations now tested and ALL falsified - local-structure loss, spectral amplification, and four further statistics (G8: D of 0.62-0.80 vs transfer's 4.03). The cliff is real, reproduced and UNEXPLAINED

Series H / V — certification

check	result
H1 artifact battery	47 checks, NO hard failures; 2 flags, both the documented GPT-2 collapse, reproduced independently in two files
rows per input dimension	every fit 5.9-9.8, above the pre-registered floor of 5
alignment (shuffle test)	honest fits beat shuffled by 0.545-0.863 vs a 0.2 threshold
V1 independent reproduction	verification AUC 0.9994, R@1 0.987, 69x knob spread - three reported figures from a separate code path

Series G / K - addenda, 23 August 2026

experiment	result
G0 hub reconstruction	original G-series artefacts were never persisted; hub rebuilt from the cached spaces and GATED on the published spectrum. PASS at 0.71% shape error (widths 128-768); the 0.846 global scale is cancelled exactly by whitening. The head protocol did NOT survive - see G4-calib
G4-calib protocol control	SigLIP 2 through the REBUILT pipeline scores 86.4% against its published 94.2%, so the reconstruction runs 7.8 pts low. Cause identified: the original hub was 4 spaces, per-space scaled, random split, bge targets, alpha 1e-2
G4-conv cross-ARCHITECTURE	ConvNeXt-base (convnet, IN-22k) 87.5% vs SigLIP 86.4% measured identically - a convnet sits ABOVE an in-band encoder. Architecture does NOT bound the claim; architecture and objective remain confounded. Row alignment tested (separation 0.418), control 0.000
G8 four more cliff statistics	accumulated noise energy 0.62, noise/signal ratio 0.80, condition number 0.68, shared-direction fraction 0.72 - against transfer's 4.03 on a pre-registered discontinuity standard. ALL FOUR FAIL. Six explanations tested, six falsified; the cliff is still UNEXPLAINED
K1 raw-space Procrustes	ORDERING reproduces with no PCA step anywhere: bge-SBERT +0.738 top, GPT-2 pairs 0.059-0.069 bottom - Table C12k is not a reduction artefact. A similarity transform reaches 0.281 vs ridge's 0.667 (42%). Levels NOT comparable to the rotation-only 0.038 / 0.194
K2 G7 re-run	stratification reproduces on the rebuilt hub: GPT-2 preservation 0.262 (pub 0.257), Spearman +0.891 (pub +0.820), degenerate pairs +0.122 (pub +0.151). SBERT again healthy at 0.135
K3 the null, bounded	cluster-bootstrap 95% CI -0.038 to +0.024, MDE 0.034 against a pre-registered SESOI of 0.050 - a DEMONSTRATED null. Six pairs from four encoders, so encoders are resampled, not pairs (1.22x wider). Rebuilt -0.010 vs original +0.010: no stable sign

Nine findings, eight explanations tested and recorded as FALSE (two of them the project's own), and a theory that survived its own out-of-sample test. Every number here is held-out and quoted against a control. The central local-vs-global result was later checked against a calibration standard proposed independently after the analysis was designed, and passed. The addenda above ran on a hub rebuilt from surviving caches; how that reconstruction differs from the original is stated in Appendix E rather than smoothed over.